Doan Du

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EDUCATION

University of Missouri-Kansas City (UMKC) — **Kansas City, MO**
Master of Science in Computer Science — *Aug 2025 - Expected May 2027*
Coursework: Principles of Big Data Management, Big Data Analytics, Advanced Software Engineering, Cloud Computing, Deep Learning, Principles of Data Science, Machine Learning

University of Greenwich — **Ho Chi Minh City, Vietnam**
Bachelor of Science in Computer Science — *Jan 2019 - Jan 2023*

TECHNICAL SKILLS

Languages: Python, SQL, TypeScript, JavaScript (ES6+), Bash, HCL (Terraform), HTML5, CSS3
Data Engineering & Streaming: Apache Spark 3.5 (PySpark, Spark SQL, Structured Streaming, MLlib), Apache Kafka, Snowflake, Apache Parquet (Medallion Lakehouse), Pandas, PyArrow
AI/ML & LLM Systems: PyTorch, HuggingFace (Transformers, PEFT, Unsloth), Llama-3 QLoRA, Google Gemini SDK, Hybrid RAG (FAISS, BM25, CrossEncoder), Scikit-learn, XGBoost, Optuna, Conformal Prediction
Backend, Frontend & Optimization: FastAPI, React 18, Next.js 14, Google OR-Tools (CP-SAT), SQLModel, Pydantic v2, Streamlit
Cloud, Databases & DevOps: AWS (WAF v2, API Gateway, Kinesis, Lambda, DynamoDB, S3, CloudWatch, SNS, Amplify), PostgreSQL (Supabase), MySQL, SQLite, Docker, GitHub Actions CI/CD, Pytest, Vitest, Git

ENGINEERING PROJECTS

HyperLogistics-Stream: Real-Time Flight & Freight Delay Lakehouse — **Oct 2026 - Nov 2026**
PySpark 3.5, Kafka, Parquet, FastAPI, Docker — github.com/JoeDoan/HyperLogistic_Stream
- Engineered a Bronze-Silver-Gold Parquet medallion lakehouse across 40 US passenger and freight hubs (FedEx, UPS, Amazon Air) over 20.55 GB of BTS On-Time and NOAA weather records (14.38M train and 3.04M 2024 holdout flights).
- Built a warm-JVM Spark MLlib GBTRegressor hyperparameter hillclimb pipeline with 14-feature weather ablation, reaching 9.77 min validation MAE (18.8% lower than baseline) on cached Parquet partitions.
- Enforced UTC session pinning, midnight-rollover date guards, Haversine aircraft deduplication, and Spark contract tests on a Kafka Structured Streaming pipeline sustaining 4,448 records/sec across 48 benchmark batches.

OptiSched: Conformal Enrollment Forecasting & CP-SAT Scheduler — **Sep 2026 - Nov 2026**
FastAPI, OR-Tools, React 18, TypeScript — github.com/JoeDoan/CS5551_Advanced-Software-Engineering-
- Authored a Google OR-Tools CP-SAT course-timetabling solver with domain pruning (>70% search space reduction, 0 hard conflicts), JWT RBAC, 3-tier instructor preference precedence, and React calendar/matrix workspaces.
- Built a 10-year UIUC FOIA registrar dataset pipeline (22,991 sections across 40 STEM courses, 2016 to 2026) and held-out Spring 2026 replay benchmark, cutting unseated students by 61% vs. point forecasts (82% vs. prior-term copy).
- Built a Commercialization Advisor v2 scope gate, readiness coach, and evaluation suite over 425 BM25-indexed policy passages (100% hit@5, 1.0 MRR across 25 scenarios), backed by 416 CI tests (256 Pytest, 160 Vitest).

LexGuard: Legal Contract Compliance & Hybrid RAG System — **Feb 2026 - Apr 2026**
Python, Snowflake, Llama-3 QLoRA, Gemini 2.5, Docker — github.com/JoeDoan/Big_Data_Project
- Built an OCR-fallback PDF-to-Snowflake ingestion pipeline (VARIANT JSON chunks) and multi-table chat session persistence with source-citation tracking for automated CUAD commercial contract risk auditing.
- Fine-tuned and hosted 4-bit Llama-3 8B QLoRA (Unsloth/PEFT) and Legal-BERT adapters on CUAD clauses, fixing chat-template token loops across a hybrid FAISS, BM25, and CrossEncoder retrieval agent.
- Implemented an end-to-end LLM evaluation harness across CUAD test contracts, raising risk-clause and metadata extraction accuracy from 53.8% (512-token BERT QA) to 86.3% via 200K-character Gemini 2.5 Flash extraction.

Cloud-Native SQL Injection Detection & Telemetry Pipeline — **Apr 2026 - May 2026**
Terraform, AWS (WAF, Kinesis, Lambda, DynamoDB), React — github.com/JoeDoan/Cloud_project
- Authored Terraform IaC modules for AWS WAF v2, throttled REST API Gateway, CloudWatch-to-Kinesis Firehose streaming, AES-256 S3 Glacier archival, DynamoDB, SNS, and Amplify within a 14-service stack (87% CIS v1.4 score).
- Upgraded the Python Lambda log classifier with URL-percent decoding and gzipped CloudWatch log unwrapping to catch the 6% of encoded SQLMap payloads bypassing WAF (94% WAF-blocked), writing to DynamoDB in <2s with SNS alerts.

Non-Custodial Crypto Circuit-Breaker DCA & Quant Engine — **Apr 2026 - Oct 2026**
Python, XGBoost, Optuna, Pandas, GitHub Actions
- Designed a pure state machine (ACCUMULATE, PROTECT, DEPLOY) shared between 9-year historical backtests (2017 to 2026, 24 Pytest replay cases) and hourly GitHub Actions runs polling CoinMetrics MVRV Z-Score and FRED macro feeds.
- Developed a walk-forward 10-seed bagged XGBoost and Optuna research pipeline with High-Water Mark risk sizing on daily candles, reducing cross-seed return variance from 11.17% to 1.43%.

WORK EXPERIENCE

Alta Media — **Ho Chi Minh City, Vietnam**
Front-end Developer Intern | github.com/DoanDu/QueuingSystem — *Feb 2023 - Apr 2023*
- Developed a web-based queuing and kiosk administration dashboard in ReactJS and TypeScript for real-time ticket tracking, service-counter monitoring, and operational reporting.
- Implemented user authentication and role-based access control (RBAC) guards to scope staff permissions across counter and administrative views, participating in Git code reviews.